



Thin liquid film method for analyzing gas–liquid annular flow in nonstraight pipe components

Ri Zhang^{*}, Mengyan Ding

College of Engineering, Ocean University of China, Qingdao 266100, China

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ABSTRACT

This study employs the thin liquid film method (TLFM) to analyze annular flow in nonstraight pipe components. The TLFM considers the interaction among the liquid film, gas phase, and droplets and requires a specific numerical method for implementation. Following thorough verification against experimental results from straight pipelines, the TLFM is applied to investigate the evolution of annular flow in a U-bend, a 90-degree bend, and a T-junction. In bends, the flow field morphology is dominated by two symmetrical cyclones generated by the gas phase. In the T-junction, gravity significantly influences annular flow formation in the main branch, while the gas cyclone causes redistribution of the liquid phase within the liquid film in the side branch.

1. Introduction

The gas–liquid annular flow is a common multiphase flow pattern in nuclear reactors, wherein part of the liquid phase adheres to the inner wall of the pipeline in a liquid film form and moves along the pipeline, and part of the same phase exists in the gas core in droplet form and moves with the gas phase. The intense momentum and mass exchange between the liquid film and the gas core shapes the annular flow field. The flow field undergoes more complex changes when the gas–liquid annular flow enters the nonstraight pipe components (e.g., elbows and T-junctions).

Many classical mechanistic models were developed to analyze the gas–liquid annular flow in long straight pipes. Wallis' work (1969) formed the basic rudiments of subsequent models, separately analyzing the flow field in the liquid film and gas core. The pressure gradient and the shear stress in the liquid film and gas core were derived based on the stress distribution. Hewitt and Hall-Taylor (1970) established the basic equations of the gas–liquid annular flow and proposed the famous triangular relationship stating that the flow rate, shear stress, and liquid film thickness affect and restrict each other in annular flow. Govier and Aziz (1972) deduced the velocity distribution in the liquid film of annular flow based on the velocity distribution in a single-phase pipe flow. They also calculated the corresponding pressure gradient in the liquid film and gas core. Whalley (1987) developed the triangular relationship and provided a theoretical solution for the liquid film flow rate. The mechanistic model construction generally requires the usage of

supplementary equations for the entrainment ratio (Schadel et al., 1990; Lopez de Bertodano et al., 1997; Kataoka et al., 2000; Sawant et al., 2008; Cioncolini and Thome, 2012), liquid film thickness (Henstock and Hanratty, 1976; Fukano and Furukawa, 1998; Liu et al., 2024), shear stress (Henstock and Hanratty, 1976; Fukano and Furukawa, 1998; Ishii and Mishima, 1984; Fore et al., 2000), droplet size (Tattersson et al., 1977; Kataoka et al., 1983; Ambrosini et al., 1991), etc. However, these mechanistic models and supplementary equations can be applied only to long straight pipelines.

Usually, the annular flow in nonstraight pipe components is experimentally or numerically studied with the aid of computational fluid dynamics (CFD). Ribeiro et al. (2001) measured the Sauter mean size of the annular flow droplets before and after a 90° horizontal bend. The experimental results showed that the droplet became larger after passing through the bend. Conte and Azzopardi (2003) experimentally studied the division of the gas–liquid two-phase flow at a large-diameter T-junction and found that the liquid phase dominated the phase split of the annular flow at the T-junction. Abdulkadir et al. (2014) measured the liquid film thickness distribution in a 180° return bend. The experimental results depicted that the liquid film inside the bend was thicker due to the action of gravity. Meanwhile, Aliyu et al. (2017) performed an experiment to study the influence of the U-bend on the downstream gas–liquid two-phase flow. They fitted an empirical formula of the liquid film thickness in large-diameter pipes based on the liquid film thickness measured at different positions downstream of the bend. Li et al. (2017) experimentally researched the influence of the superficial velocity, surface tension and liquid viscosity on the phase split of the gas–liquid

^{*} Corresponding author.

E-mail address: zhangri@ouc.edu.cn (R. Zhang).

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Nomenclature

C_D	Drag coefficient (-)
D	Pipe diameter (m)
$\bar{d}_d$	Average droplet size (m)
d_{32}	Sauter mean diameter (m)
E	Entrainment ratio (-)
$\vec{g}$	Gravity (m/s^2)
J	Superficial velocity (m/s)
k_s	Equivalent roughness (m)
M_{di}	Parcel mass (kg)
$\dot{m}_d$	Mass sources ($\text{kg/m}^2/\text{s}$)
$\dot{m}_d^D$	Deposition rate ($\text{kg/m}^2/\text{s}$)
P	Pressure (Pa)
$\vec{q}_d$	Momentum sources (kg/m/s^2)
Re_{lf}	Liquid film Reynolds number (-)
R_λ	Wave birth rate (s^{-1})

$\vec{S}_M$	Mass sources ($\text{kg/m}^2/\text{s}$)
Δs	Grid area (m^2)
Δt	Time step (s)
$W_{l/\text{Inject}}$	Mass flux ($\text{kg/m}^2/\text{s}$)

Greek letters

α	Fitted parameter (-)
β	Sheltering parameter (-)
δ	Film thickness (m)
$\bar{\lambda}$	Average wavelength (m)
μ	Viscosity (Pa·s)
ρ	Density (kg/m^3)
σ	Surface tension (N/m)
$\vec{\tau}_{fg}$	Interfacial shear stresses (Pa)
$\vec{\tau}_{fw}$	Wall shear stresses (Pa)
ϕ	Polar angle (°)
φ	Azimuth angle (°)

two-phase flow in a micro-T-junction. Their test showed that the liquid taken off at the T-junction was dominant for the annular flow evolution. Liu et al. (2022) provided direct insights into the physical triggering mechanism of critical heat flux (CHF) through a visualization experiment. They observed a three-layer structure close to CHF conditions, comprising the liquid film, vapor layer, and liquid core. The above-mentioned experiments depict that annular flow undergoes a complex evolution after entering the nonstraight pipe components.

The numerical procedure relying on CFD generally analyzes annular flow in two ways. The first approach (Han and Gabriel, 2007; Ghosh et al., 2011; Liu et al., 2011; Jiang et al., 2019; Kiran et al., 2020; Wu et al., 2020; Zahedi and Rad, 2022) uses the volume of fluid (VOF) model to describe the immiscible liquid and gas phases interpenetrating each other in the pipeline. The VOF model can solve the volume ratios of the liquid and gas phases in a mesh. The interface between the liquid and gas phases exists in a mesh containing both liquid and gas phases. However, the interface determination requires the application of geometric or mathematical reconstruction schemes, which makes the VOF model unable to directly identify the liquid film. The second approach separately tracks the liquid film adhering to the inner wall of the pipeline using the film's governing equation. This approach is referred to herein as the liquid film construction method. The gas phase and the droplets in the gas core are studied using Eulerian–Lagrangian mode (Adechy and Issa, 2004; Li and Anglart, 2019) or Eulerian–Eulerian mode (Alipchenkov et al., 2004; Kishore and Jayanti, 2004; Guan et al., 2015; Li and Anglart, 2016). The key to this approach is to describe the mass and momentum exchange between the gas core and the liquid film. The empirical or semiempirical relationships fitted from the experiments in straight pipes are usually employed as the governing equations of the mass and momentum exchange.

The numerical analysis of the annular flow in the nonstraight pipe components covers the two abovementioned approaches. Adechy and Issa (2004) modeled the annular flow through a T-junction using the liquid film construction method. They developed transport and momentum equations for liquid films and applied the Eulerian–Lagrangian mode to analyze the gas phase and the droplets. The velocity distribution in the liquid film was assumed to be one-seventh power profile. The empirical expression of the entrainment rates fitted in the horizontal straight pipes was used to reflect the mass and momentum exchange between the gas core and the liquid film. Ghosh et al. (2011) used the VOF model to simulate the annular flow in the return bends. The procedure was verified by comparing the numerical and experimental velocity contours through a U-bend. The numerical results showed that the pressure reduction effect of the core flow diminished as the curvature

radius increased. Wu et al. (2020) introduced the VOF model to investigate the hydrodynamic characteristics of the annular flow in a 90° elbow. The numerical research discussed the influence of the fluid properties on the annular flow morphology and the transportation efficiency in the bend.

In summary, studies on the annular flow in the nonstraight pipe components, apart from experimental research, often involves the use of the VOF and liquid film construction methods. The VOF method faces challenges in directly identifying the interface between the liquid film and the gas core. Meanwhile, the liquid film construction method relies on relationships fitted in straight pipes, which is not suitable for the complex flow filed boundaries in nonstraight pipe components. Our previous work (Zhang et al., 2024) proposed a novel thin film method (TLFM) for analyzing annular flow, specifically addressing and overcoming the limitations of the VOF and liquid film construction method. The TLFM is uniquely suited for nonstraight pipe components, offering the capacity to identify the gas–liquid interface and analyze the momentum and mass exchange between the phases at a microscopic scale. Building on this method, the present work provides a detailed discussion on the evolution of the morphological and dynamical properties of annular flow in nonstraight pipe components.

2. Basic theory and numerical methods

In straight pipelines, Kelvin–Helmholtz waves are generated on the surface of the liquid film due to the gas phase disturbance. Some liquid may be stripped off from the liquid film surface to form droplets during the Kelvin–Helmholtz wave propagation, which is the atomization process of annular flow. Due to the gas core turbulence, these droplets move irregularly along the pipeline and may deposit on and integrate into the liquid film, depicting the deposition process of annular flow. When the deposition and atomization processes reach a dynamic equilibrium, annular flow is fully developed in the straight pipelines. However, this equilibrium is disrupted when the liquid and gas phases enter the nonstraight pipe components. In such cases, droplets collide with the liquid film due to their inertia. Furthermore, the abrupt change in flow direction leads to the redistribution of the liquid film by the combined effects of gravity and the gas phase.

The TLFM separately studies the liquid film and the gas core, while considering the interaction among the liquid film, droplets, and gas phase. It is comprised of a basic theory and a corresponding numerical method.

2.1. Liquid film construction

2.1.1. Basic theory

The liquid film thickness δ and the film velocity $\vec{v}_l$ are the key parameters of the liquid film adhering to the inner wall of the pipeline. The governing equations are as follows (Zhang et al., 2024):

$$\frac{\partial \rho_l \delta}{\partial t} + \nabla_s \cdot (\rho_l \delta \vec{v}_l) = \dot{m}_d \quad (1a)$$

$$\begin{aligned} \frac{\partial \rho_l \delta \vec{v}_l}{\partial t} + \nabla_s \cdot \rho_l \delta \vec{v}_l \vec{v}_l = \\ -\delta \nabla_s [P - \rho_l \delta \vec{n} \cdot \vec{g} - \sigma \nabla_s \cdot (\nabla_s \delta)] + \rho_l \delta (\vec{g} \cdot \vec{t}) \vec{t} + \vec{\tau}_{fg} - \vec{\tau}_{fw} + \vec{q}_d \end{aligned} \quad (1b)$$

where ρ_l is the liquid density; σ is the surface tension; P is pressure; $\vec{g}$ is gravity; $\vec{n}$ and $\vec{t}$ are the unit vectors perpendicular and parallel to the pipe wall, respectively; ∇_s is the gradient along the pipe wall; $\dot{m}_d$ and $\vec{q}_d$ are the mass and momentum sources, respectively, which are discussed in Section 2.2; and $\vec{\tau}_{fg}$ and $\vec{\tau}_{fw}$ are the interfacial and wall shear stresses, respectively.

The interfacial shear stress $\vec{\tau}_{fg}$ represents the liquid film resistance on the gas phase, while the wall shear stress $\vec{\tau}_{fw}$ represents the pipe wall resistance on the liquid film. In Fig. 1, the pressure difference across the liquid film is denoted by ΔP in a pipe section of length ΔL . The stress balance equation of the liquid film is presented as follows:

$$2\pi R \Delta L \tau_{fg} - 2\pi(R+y)\Delta L \tau_l = \pi[(R+y)^2 - R^2] \Delta P \quad (2)$$

where R is the distance from the pipe axis to the liquid film surface, y is the distance of the liquid phase in the liquid film to the film surface, and τ_l is the shear stress in the liquid film. The TLFM assumes that the liquid viscosity has a dominant effect on the liquid film compared to the liquid turbulence. That is,

$$\tau_l = -\mu_l \frac{dv}{dy} \quad (3)$$

The pipe radius is much larger than the liquid film thickness (i.e., $R \gg y$). The liquid velocity is zero at the pipe wall (i.e., $v|_{y=\delta} = 0$). The velocity profile of the liquid phase is solved as follows:

$$v = \frac{1}{2} \frac{\Delta P}{\mu_l \Delta L} (y^2 - \delta^2) - \frac{\tau_{fg}}{\mu_l} (y - \delta) \quad (4)$$

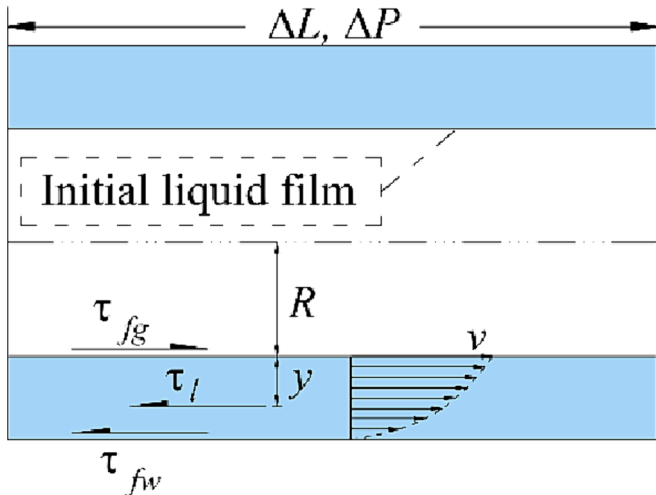


Fig. 1. Stress balance of the liquid film.

The film velocity $\vec{v}_l$ in Eq. (1) is actually the mean film velocity; hence, the following expression can be obtained:

$$v_l \delta = \int_0^\delta v dy \quad (5)$$

The wall shear stress is solved as

$$\tau_{fw} = \tau_l|_{y=\delta} = \frac{3\mu_l}{\delta} v_l - \frac{1}{2} \tau_{fg} \quad (6)$$

For the typical annular flow, the liquid film thickness is much smaller than the pipe diameter. Hence, the TLFM ignores the liquid film volume and treats the undulated liquid film surface as a "rough pipe wall." The equivalent roughness proposed by Adechy and Issa (2004) is employed by the TLFM as follows:

$$k_s = 7.0623\delta - 0.0088D \quad (7)$$

where D is the pipe diameter. The interfacial shear stress τ_{fg} is solved according to the equivalent roughness (Ligrani and Moffat, 1986; Ansys, 2018).

2.1.2. Numerical operations

The TLFM implementation requires some specific numerical operations. In Fig. 2, we divide the computational domain into the injection and development regions. The development region contains several observation regions. The injection region is used to inject the liquid and gas phases, the development region is for the annular flow formation, and the observation region is used to calculate the characteristic parameters of annular flow. The liquid phase is injected into the computational domain from the pipe wall of the injection region. Although the pipe wall is impermeable, the injection process can be regarded as a liquid phase entering from a "porous" wall. The liquid phase injection is governed by two parameters, that is, the mass flux $W_{l/Inject}$ and the initial liquid film velocity v_l^{init} .

The mass flux $W_{l/Inject}$ is the initial mass source $\dot{m}_d$ in Eq. (1a) determined as follows:

$$W_{l/Inject} = \frac{DJ_l \rho_l}{4L_{Inject}} = \dot{m}_d^{init} \quad (8)$$

where L_{Inject} is the injection region length affecting the liquid film thickness in the injection region. An injection region that is too short will result in an excessively thick liquid film, leading to a sharp change upon entrance to the development region and potentially causing a divergence in the numerical iteration. Conversely, an injection region that is too long will consume a relatively high amount of computational resources. After the trial calculation, the length L_{Inject} should be four to five times the pipe diameter.

In addition, the initial velocity of the liquid film along the pipeline should be set such that the liquid film can enter the development region. Similarly, to control the scale of the calculational domain and allow the numerical iteration to converge quickly, the initial liquid film velocity v_l^{init} should not differ much from the fully developed one v_l . Hence, v_l^{init} should satisfy the following relation:

$$v_l^{init} \approx (1-E)v_l + E\bar{v}_g \approx \bar{v}_g \approx J_g \quad (9)$$

where $\bar{v}_g$ is the average gas phase velocity, and E is the entrainment ratio defined as the liquid phase ratio in droplet form. The first approximation $(1-E)v_l + E\bar{v}_g$ is the average liquid phase velocity. The liquid film velocity is much smaller than the gas velocity (i.e., $v_l \ll \bar{v}_g$), and the entrainment ratio E approaches 1 in most cases; hence, the second approximation can be obtained. Because the liquid film volume is neglected, the average gas velocity $\bar{v}_g$ is approximately equal to the superficial gas velocity J_g . It should be emphasized that the initial liquid film velocity v_l^{init} only affects the numerical iteration efficiency. If the

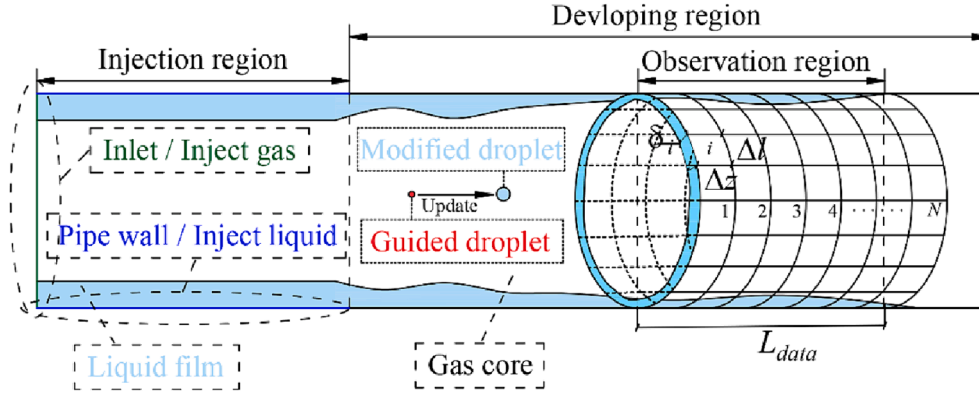


Fig. 2. Numerical operations of the thin liquid film method (TLFM).

initial liquid film velocity is too large, the liquid film will decelerate due to the wall shear stress τ_{fw} . By contrast, if the initial liquid film velocity is too small, the liquid film will accelerate due to the interfacial shear stress τ_{fg} . The initial momentum source $\vec{q}_d$ in Eq. (1b) is expressed as follows:

$$\vec{q}_d^{init} = W_{l/inject} \vec{v}_l^{init} \quad (10)$$

Fig. 3 depicts the numerical operations for the liquid film construction.

2.2. Droplet tracking

2.2.1. Basic theory

Droplets are formed during the atomization process and eliminated during deposition. In the atomization process, a Kelvin–Helmholtz wave appears on the liquid film surface due to the relative velocity between the liquid and gas phases. The liquid phase then peels off from the Kelvin–Helmholtz wave and forms droplets. Hewitt and Govan (1990) established a criterion for the atomization process initiation based on experimental results. The TLFM employs the criterion as follows:

$$Re_{if} \geq Re_{ifc} = e^{5.8504 + 0.4249 \frac{\rho_g}{\rho_l}} \sqrt{\frac{\rho_l}{\rho_g}} \quad (11a)$$

$$Re_{if} = \frac{4\delta v_l \rho_l}{\mu_l} \quad (11b)$$

where μ_g is the gas viscosity, and ρ_g is the gas density.

Next, we analyze the droplet separation process at the microscopic scale. Lopez de Bertodano et al. (1997) extended the Kelvin–Helmholtz theory through an instability analysis to determine the birth rate of the Kelvin–Helmholtz waves, allowing the liquid phase separation. The birth rate R_λ , defined as the number of breaking Kelvin–Helmholtz waves per unit time, is calculated as follows:

$$R_\lambda = 0.384 \frac{\rho_g v_{sg}^3}{\sigma} \sqrt{\frac{\rho_g}{\rho_l}} \quad (12)$$

where v_{sg} is the gas velocity at the liquid film surface.

Mayer (1961) proposed the following threshold wavelength λ_t at which the liquid phase begins to separate from the liquid film:

$$\lambda_t = 2\pi \sqrt[3]{16 \left(\frac{\mu_l \sqrt{\sigma/\rho_l}}{\beta \rho_g v_{sg}^2} \right)^{\frac{2}{3}}} \quad (13)$$

where β is the sheltering parameter recommended to be 0.3. Mayer (1961) derived the following average wavelength $\bar{\lambda}$ of the Kelvin–Helmholtz waves with wavelengths greater than the threshold λ_t :

$$\bar{\lambda} = \frac{9}{2} \lambda_t \quad (14)$$

Based on experimental data, Mayer (1961) fitted the average size of the droplets created by a Kelvin–Helmholtz wave as follows:

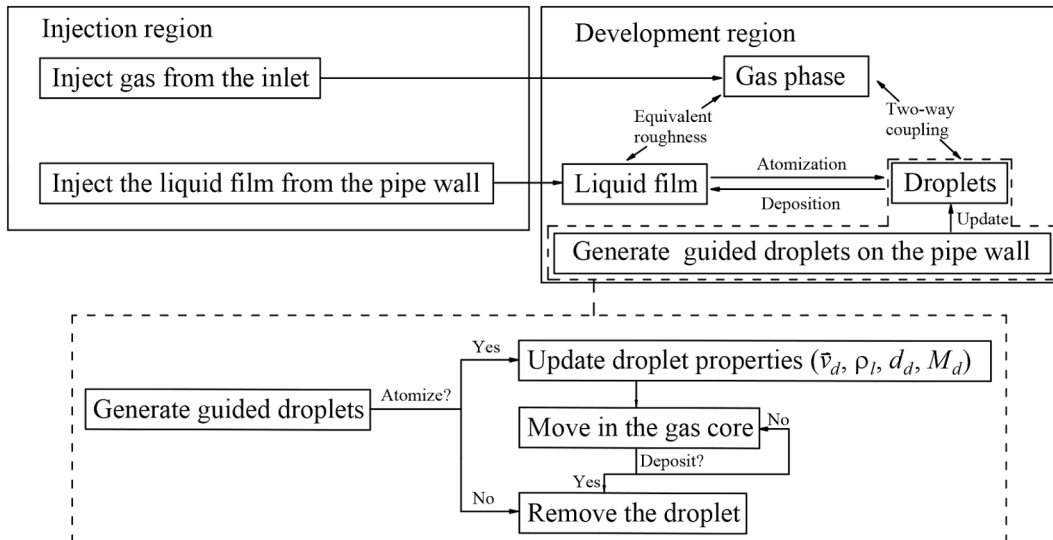


Fig. 3. Computational flow chart.

$$\bar{d}_d = \alpha \bar{\lambda} \quad (15)$$

where the fitted parameter α takes a value of 0.14. In this work, we believe that the droplet size is limited by the local liquid film thickness. That is, the average droplet size should not be larger than the local liquid film thickness:

$$\bar{d}_d = \min(\delta, \alpha \bar{\lambda}) \quad (16)$$

Eqs. (12)–(16) describe the microscopic atomization process. Let us suppose that one breaking Kelvin–Helmholtz wave produces one droplet. The mass of the liquid phase stripped off from the action area of a Kelvin–Helmholtz wave is then presented as $\pi \bar{d}_d^3 \rho_l / 6$. Considering all the breaking waves, the mass of the separated liquid phase per unit time and per unit area (i.e., atomization rate $\dot{m}_d^A$) is

$$\dot{m}_d^A = \left(\frac{\pi \bar{d}_d^3 \rho_l R_\lambda}{6 \lambda^2} \right) \approx \begin{cases} \frac{\alpha^3 \pi}{6} \rho_l R_\lambda \bar{\lambda}, & \alpha \bar{\lambda} \leq \delta \\ \frac{\pi}{6} \delta^3 \rho_l R_\lambda \bar{\lambda}^{-2}, & \alpha \bar{\lambda} > \delta \end{cases} \quad (17)$$

The separated droplet has the local liquid film velocity. That is, the momentum lost by the liquid film is obtained as

$$\vec{q}_d^A = \dot{m}_d^A \vec{v}_l \quad (18)$$

The gas phase affects the droplet movement after the droplet leaves the liquid film and enters the gas core. The governing equations of the droplet are as follows (Ansys, 2018):

$$\frac{d\vec{v}_d}{dt} = \vec{F}_{drag} + \vec{F}_{virtual} + \vec{F}_{pressure} + \vec{g} \quad (19a)$$

$$\vec{F}_{drag} = \frac{3C_D \rho_g}{4\rho_l d_d} \left| \vec{v}_g - \vec{v}_d \right| \left(\vec{v}_g - \vec{v}_d \right) \quad (19b)$$

$$\vec{F}_{virtual} = \frac{\rho_g}{2\rho_l} \left(\vec{v}_d \nabla \vec{v}_g - \frac{d\vec{v}_d}{dt} \right) \quad (19c)$$

$$\vec{F}_{pressure} = \frac{\rho_g}{\rho_l} \vec{v}_d \nabla \vec{v}_g \quad (19d)$$

where $\vec{v}_d$ is the droplet velocity, and C_D is the drag coefficient. The gas velocity in the governing equation is the sum of the mean and fluctuation velocities considering the influence of turbulence dispersion. The mean velocity is solved using the Navier–Stokes (N–S) equation presented in Section 2.3. The fluctuation velocity follows a Gaussian distribution with zero mean and a variance of 2/3 times the turbulent kinetic energy k (Gosman and Ioannides, 1983).

The droplets in the gas core move irregularly due to the influence of turbulence and may collide with the liquid film on the pipe wall. Stanton and Rutland (1998) demonstrated that a droplet will be completely absorbed by the liquid film when it hits the liquid film with a relatively small Weber number, which is consistent with most cases of the droplet deposition process in annular flow. Hence, the TLFM considers the droplet to be completely absorbed by the liquid film the moment the droplet trajectory reaches the pipe wall boundary.

2.2.2. Numerical operations

The droplets are created through the following steps during the atomization process: (1) a guided droplet is generated above each mesh on the pipe wall in the development region after one time step Δt ; (2) the distance between the guided droplet and the pipe wall is set to be five times the film thickness; (3) whether or not the liquid film on the pipe wall mesh is atomized is determined using Eq. (11); (4) the guided droplet is removed when Eq. (11) is not satisfied; (5) the guided droplet properties are updated when Eq. (11) is satisfied; (6) the modified

droplet material is set to be the liquid phase; (7) the modified droplet size is set using Eq. (16); (8) the initial droplet velocity is set as the velocity of the corresponding liquid film; (9) the droplet motion state and position are updated using Eq. (19) in one time step; and (10) steps (1)–(9) are repeated.

The guided droplets must be set up to create discrete phases in the computational domain. The guided droplet is either removed or replaced by the modified droplet; thus, only the initial position of the guided droplet must be properly set. Wallis (1969) believed that the average wave height of the breaking Kelvin–Helmholtz wave is four times the liquid film thickness. We apply this conclusion to step (2), where the guided droplet distance from the pipe wall is 5δ . The droplet generated in the numerical procedure is not a real droplet but a sample representing a group of droplets with similar properties, i.e., a parcel. The parcel mass M_{di} in each modified droplet i is recorded as follows:

$$M_{di} = \dot{m}_d^A \Delta s^A \Delta t \quad (20)$$

where Δs^A is the pipe wall mesh area where the liquid film is atomized.

The numerical operations in the deposition process are relatively simple. As the name suggests, the basic assumption of the TLFM is to ignore the liquid film volume. The liquid film thickness is the only data recorded on the pipe wall mesh. Once a droplet hits the pipe wall, it is removed from the computational domain, and the momentum and mass recorded in the droplet are added to the local liquid film. If N_d droplets are deposited in the pipe wall mesh Δs^D in one time step Δt , the deposition rate $\dot{m}_d^D$, defined as the mass of the deposited liquid phase per unit area and per unit time, is obtained as follows:

$$\dot{m}_d^D = \sum_{i=1}^{N_d} \frac{M_{di}}{\Delta s^D \Delta t} \quad (21)$$

The momentum obtained by the liquid film is represented as

$$\vec{q}_d^D = \sum_{i=1}^{N_d} \frac{M_{di} \vec{v}_{di}}{\Delta s^D \Delta t} \quad (22)$$

The mass and momentum sources in Eq. (1) are affected by the atomization and deposition processes. That is,

$$\dot{m}_d = \dot{m}_d^D - \dot{m}_d^A \quad (23a)$$

$$\vec{q}_d = \vec{q}_d^D - \vec{q}_d^A \quad (23b)$$

These processes are implemented only in the development region for droplets, as they are suppressed in the injection region. Fig. 3 illustrates the numerical operations for tracking droplets.

2.3. Gas phase modeling

The gas phase in the gas core is modeled herein using the conventional method. The governing equations of the gas phase are the N–S equations:

$$\frac{\partial \rho_g}{\partial t} + \nabla \cdot (\rho_g \vec{v}_g) = 0 \quad (24a)$$

$$\begin{aligned} \frac{\partial (\rho_g \vec{v}_g)}{\partial t} + \nabla \cdot (\rho_g \vec{v}_g \vec{v}_g) &= \rho_g \vec{g} - \nabla P + \mu_g \nabla \cdot \left[\nabla \vec{v}_g + \left(\nabla \vec{v}_g \right)^T \right. \\ &\quad \left. - \frac{2}{3} \nabla \cdot \vec{v}_g \tilde{I} \right] + \vec{S}_M \end{aligned} \quad (24b)$$

where $\vec{v}_g$ is the gas velocity, $\tilde{I}$ is the identity tensor, and $\vec{S}_M$ is the source term representing the droplet effects on the gas phase.

In this work, the standard k – ϵ turbulence model is used to describe the Reynolds stress and solve the N–S equations. The interaction

between the gas phase and the droplets is reflected using the two-way coupling method. The gas phase enters the computational domain from the inlet of the injection region (Fig. 2). At the boundary, the gas phase contacts the liquid film rather than the pipe wall. The equivalent liquid film roughness is considered in Eq. (7).

3. Validation and discussion

Although the TLFM is designed to analyze the gas–liquid annular flow in nonstraight pipe components, it should also be applicable to simple long straight pipelines. In addition, the experimental data of the annular flow in long straight pipes are sufficient. Hence, we chose the characteristic parameters of the annular flow in long straight pipes to verify the TLFM. The experimental data are selected from five tests and include film thickness δ , entrainment ratio E , deposition rate $\dot{m}_d^D$, pressure gradient ΔP , wall shear stress τ_{fw} , and Sauter mean diameter d_{32} .

3.1. Experimental parameters

Lopez de Bertodano et al. (1997) performed an experiment in an upward flow loop to measure the entrainment ratio E and the deposition rate $\dot{m}_d^D$ in annular flow. The two extractions installed in the loop were used to remove the liquid film and the deposited liquid phase. The deposited liquid phase removed by the second extraction can be converted into the deposition rate. Wolf et al. (2001) measured the pressure gradient ΔP and the wall shear stress τ_{fw} in a vertical annular flow. The pressure drop over the test section was measured by employing a differential pressure transducer. The wall shear stress was measured using a hot film probe. Schubring et al., (2010a,2010b) performed planar laser-induced fluorescence measurements to observe the liquid film in annular flow. The laser images were analyzed to obtain the liquid film thickness δ data. The entrainment ratio E was calculated based on the liquid film flow rate. Alamu and Azzopardi (2011) designed an experiment that measured the film thickness δ and the pressure gradient ΔP of annular flow. The film thickness measurement was performed using a conductivity probe consisting of a pair of ring electrodes. The pressure drop across the test section was monitored using differential pressure transducers. Wang et al. (2020) employed phase Doppler anemometry to measure the size of droplets in annular flow and calculated the Sauter mean diameter d_{32} of the droplets based on the measured data. Table 1 lists the experimental parameters.

3.2. Computational domain and meshes

We model 46 cases from the five annular flow tests in straight pipes. The computational domain shape and the corresponding meshes are almost the same in each case. Fig. 4(a) shows the computational domain meshed using structured grids. The development region contains an

observation region to calculate the characteristic parameters of annular flow (Fig. 2). The observation region length L_{data} is approximately 10 times the pipe diameter. The distance between the observation region and the inlet is 40 times the pipe diameter to ensure that annular flow is fully developed.

The liquid film thickness in the observation region is treated as the target variable to verify the mesh independence. The liquid film thickness δ fluctuates within a range instead of maintaining a constant value. Hence, we take the average of the liquid film thickness over a period as the liquid film thickness. The period needs to reach 2 s, during which the fluctuation of the liquid film thickness does not exceed 10 %. We randomly take one case from each test as an example to show the liquid film thickness δ determination (Fig. 5).

The computational domain is meshed using different numbers of cells to analyze the mesh sensitivity, as shown in Fig. 6. The number of cells ranges from 100,548 to 564,480. Taking Case 1 as an example, the predicted liquid film thickness δ_p does not change significantly when the number of cells increases from 353,780 to 564,480. Considering both accuracy and computational efficiency, the present work selects the computational domain with Mesh 3.

3.3. Validation of the TLFM

Except for the entrainment ratio E , all characteristic parameters can be solved directly. The calculation method of the entrainment ratio can be found in our previous work (Zhang et al., 2024). Fig. 7 compares the measured and predicted results of the six characteristic parameters. The ordinate and abscissa represent the predicted and experimental results, respectively. The solid lines indicate that the predicted value exactly matches the measured value. The closer the data points are to the solid line, the more precise the predictions become. The corresponding figure lists the mean absolute percentage error (MAPE) of each characteristic parameter. The data points representing the deposition rate $\dot{m}_d^D$ are closely adjacent to the solid line. Those representing the film thickness δ , pressure gradient ΔP , wall shear stress τ_{fw} , and Sauter mean diameter d_{32} are evenly overlaid on the solid lines. The prediction results of the five abovementioned characteristic parameters of annular flow agree well with the measured ones. However, overall, the data points of the entrainment ratio E are on the upper side of the solid line, indicating that the predicted results are greater than the measured ones.

The MAPE of the entrainment ratio E reaches 43.15 %, which is significantly larger than those of the other parameters. One possible reason for the overestimation of the entrainment ratio E is the assumption made by the TLFM that the liquid film comprises only the liquid phase, neglecting the gas phase. Both the predicted and measured entrainment ratios E are calculated based on the liquid film parameters. In reality, the liquid film in annular flow is mixed with the gas phase. Neglecting the gas phase in the liquid film can lead to an

Table 1
Experimental parameters.

Experiment	No. of cases	Test parameters	D (mm)	J_l (m/s)	J_g (m/s)	R/D	P (bar)	Fluid
Maddock et al. (1974)	1	δ	25.4	0.06	60.62	10	1	Air–water
Lopez de Bertodano et al. (1997)	6	$E, \dot{m}_d^D$	9.5	0.089	30–80	–	2.03	Air–water
Wolf et al. (2001)	9	$\Delta P, \tau_{fw}$	31.8	0.01–0.12	52.82	–	2.38	Air–water
Conte and Azzopardi (2003)	1	δ	127	0.28	24.5	–	1	Air–water
Schubring et al., (2010a,2010b)	8	δ, E	23.4	0.063	40–100	–	1	Air–water
Alamu and Azzopardi (2011)	7	$\delta, \Delta P$	19	0.05	30–45	–	1.5	Air–water
Abdulkadir et al. (2014)	1	δ	127	0.02	16.2	3	3	Air–water
Wang et al. (2020)	16	d_{32}	50	0.04–0.079	30–60	–	1	Air–water

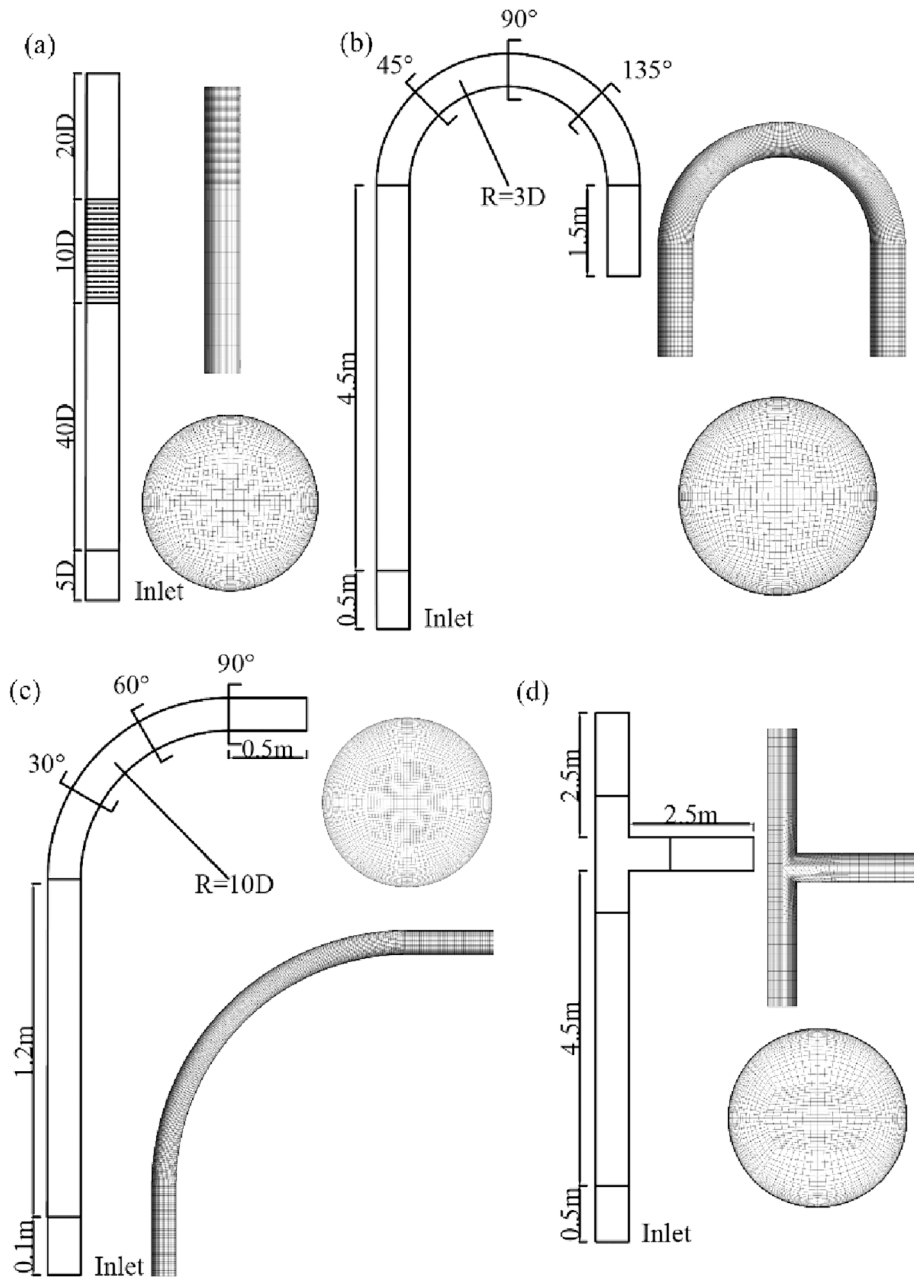


Fig. 4. Computational domain and mesh: (a) straight pipe, 353,780 cells; (b) U-bend, 657,020 cells; (c) 90° bend, 388,864 cells; and (d) T-junction, 521,424 cells.

underestimation of the film thickness δ , which in turn distorts the calculated entrainment ratio E .

The TLFM validation introduces six characteristic parameters of the annular flow selected from five different experiments and involves 81 data points from 46 cases, ensuring that the TLFM is fully verified.

3.4. Applicability of the TLFM

The basic assumption of the TLFM is to neglect the liquid film volume, which is reasonable for a typical annular flow. In Fig. 8, the experimental data used herein to verify the TLFM are marked on a classical multiphase flow regime diagram (Barnea, 1987). All data points are within the annular flow range in the diagram, where the liquid film thickness is significantly less than the pipe diameter. The TLFM is generally applicable when the liquid film thickness in annular flow is less than 1/20 of the pipe diameter.

4. TLFM application

The TLFM is used herein to analyze the evolution of annular flow entering nonstraight pipe components, including the U-bend, 90° bend, and T-junction.

4.1. Annular flow in a U-bend

The flow loop established by Abdulkadir et al. (2014) contained a transparent U-bend that could be used to observe the flow field change of the annular flow entering the nonstraight pipe components. The liquid film thickness in the U-bend was measured using a parallel wire probes and flush-mounted pin. The bend had a diameter of 127 mm and a curvature radius of 3 times the pipe diameter. Its turning angle was 180°. Table 1 presents the experimental parameters. In our previous work (Zhang et al., 2024), we applied the TLFM to calculate the liquid film thickness of annular flow in the U-bend. As shown in Fig. 4(b), the

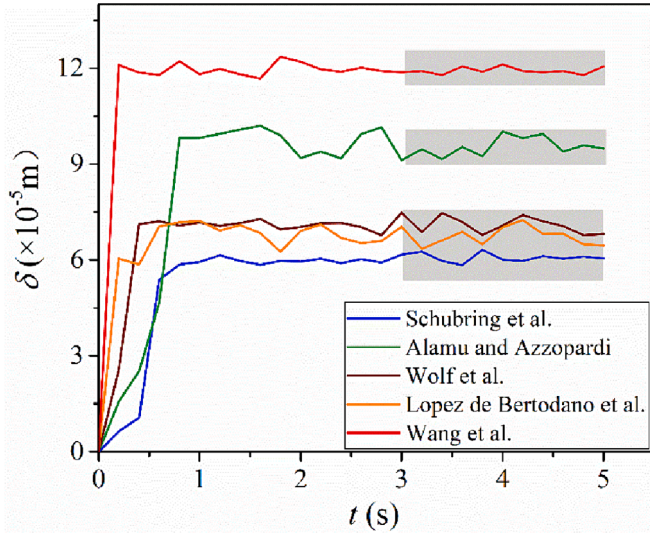


Fig. 5. Determination of the liquid film thickness δ .

computational domain was meshed using structured grids and divided into the injection and development regions.

The comparison between the predicted and measured liquid film thicknesses at different bend sections is shown in Fig. 9. The polar coordinates in the figure represent the three cross-sections in the bend, with angle φ marked at the center as the azimuth of the cross-section. The polar angle $\phi = 0^\circ$ corresponds to the outside of the bend, while $\phi = 180^\circ$ corresponds to the inside of the bend. The blue solid lines and the red points represent the predicted and experimental results, respectively. We made the comparison clearer by magnifying both the experimental and predicted liquid film thicknesses by a factor of 10. The MAPE of the predicted liquid film thickness is 19.82 %, revealing a good agreement between the measured and predicted results.

Fig. 10 depicts the evolution of the annular flow entering the U-bend. The three circles denote the cross-sections of the pipes at their corresponding positions. The black arrow lines on the cross-sections and the red solid lines along the bend represent the streamlines of the gas phase in the gas core and the liquid phase in the liquid film, respectively. The blue scatter points depict the droplets near the bend's symmetry plane. The inner donut plot is the entrainment ratio E along the bend.

The gas-liquid annular flow changes after entering the bend from the vertical straight section due to three effects. First, droplets in the gas core hit and blend into the liquid film on the outside of the bend due to the action of inertia. Simultaneously, the liquid film on the inside of the bend sharply thickens, intensifying the local atomization process. However, the droplets generated on the inside of the bend rapidly move downstream of the bend and fail to fill the gaps of the droplets hitting the outside of the bend. Hence, two clear droplet voids appear in the gas core (Fig. 10), and the corresponding entrainment ratio E significantly

decreases. Although numerous droplets are deposited on the outside of the bend, the local liquid film is not thicker than those in the other locations. The other two effects can explain this phenomenon.

Second, the U-bend is placed vertically. The morphology of the liquid film entering the U-bend deforms due to the action of gravity. The liquid phase in the liquid film on the outside of the bend flows to the inside of the bend. The liquid phase incorporated into the liquid film during the droplet deposition process is then transported away. However, the effect of gravity on the liquid film redistribution is not dominant. We recalculate the liquid film thickness in the U-bend without considering gravity. Fig. 9 presents the predicted liquid film thickness through the green dotted lines. The influence of gravity on the liquid film is not obvious on sections with azimuth angles φ of 135° and 45° . On the section where φ is 90° , gravity only affects the liquid phase accumulation range at the pipe bottom.

Third, the cyclones created by the gas phase reshape the annular flow in the bend. The gas phase streamlines on different bend sections form two symmetrical circular flows, which are caused by the boundary change of the flow field. That is, the gas phase motion comprises two symmetrical cyclones. Due to the influence of the interfacial shear stress, the liquid phase motion in the liquid film shows the same tendency, causing the liquid phase to accumulate on the inside of the bend and form the thickest liquid film. Compared with gravity, the gas phase in the bend has a more significant effect on the shape of annular flow.

The TLFM presents the evolution of annular flow entering a U-bend from a vertical straight section. The gas phase enters the bend to form two symmetrical cyclones. Taking the polar angle ϕ as the coordinate, the flow directions of the two cyclones are $180^\circ-0^\circ-90^\circ-180^\circ$ and $180^\circ-0^\circ-270^\circ-180^\circ$, respectively. The liquid phase in the liquid film flows from the outside to the inside under the gas phase rotation. The two flows from the outside to the inside converge at the inside of the bend, leading to a sharp increase in the liquid film thickness. The droplets are deposited on the liquid film on the outside of the bend due to inertia and replenish the liquid phase lost in the liquid film flow from the outside to the inside. Gravity is not dominant in this process. The influence of gravity is most pronounced in the section with an azimuth angle φ of 90° , which smooths out the sharp increase in liquid film thickness on the inside of the bend.

4.2. Annular flow in a 90° bend

Maddock et al. (1974) observed the liquid film redistribution when annular flow entered a 90° bend from a vertical straight pipe section. The bend was made of Perspex with an inside diameter of 25.4 mm and a curvature of 10 times the inside diameter. The liquid film thickness in the 90° bend was measured using a needle probe mounted on a rotatable holder. Table 1 presents the experimental parameters. Our previous work (Zhang et al., 2024) calculated the liquid film thickness in the 90° bend using the TLFM. Fig. 4(c) shows the computational domain and mesh, while the comparison between the predicted and measured liquid film thicknesses at different bend sections is shown in Fig. 11. To

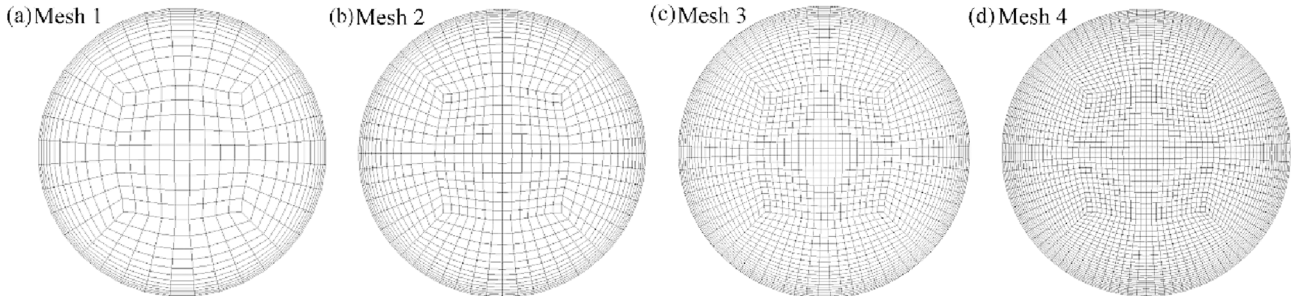


Fig. 6. Cross-sections with different cells and corresponding predicted liquid film thicknesses: (a) Mesh 1 with 100,548 cells, $\delta_p = 51.68 \mu\text{m}$; (b) Mesh 2 with 192,080 cells, $\delta_p = 57.92 \mu\text{m}$; (c) Mesh 3 with 353,780 cells, $\delta_p = 70.23 \mu\text{m}$; (d) Mesh 4 with 564,480 cells, $\delta_p = 69.66 \mu\text{m}$.

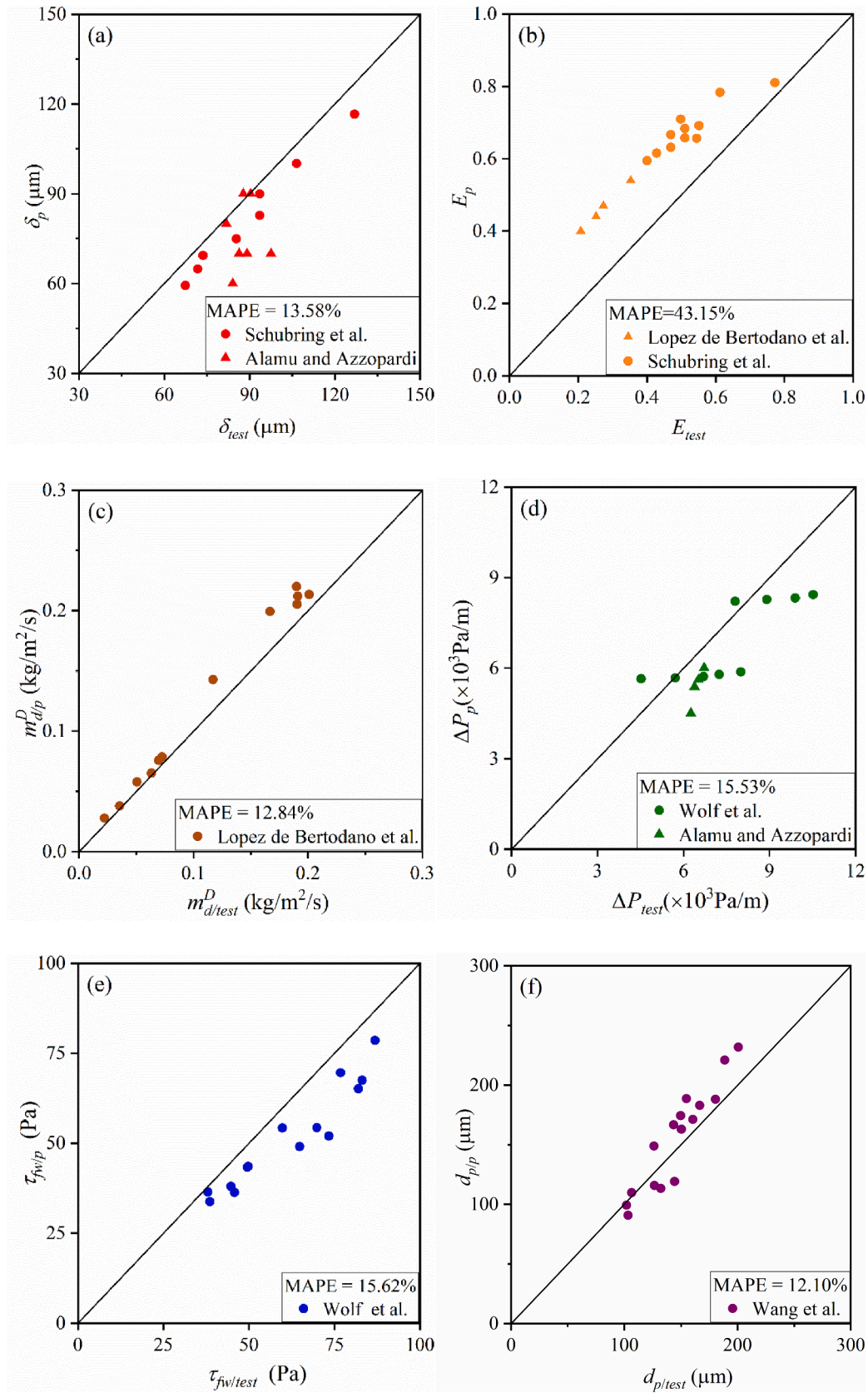


Fig. 7. Comparison of the measured and predicted characteristic parameters of the annular flow in the straight pipes: (a) liquid film thickness, δ ; (b) entrainment, E ; (c) deposition rate, $\dot{m}_{dp}^D$; (d) pressure gradient, ΔP ; (e) wall shear stress, τ_{fw} ; and (f) Sauter mean diameter, d_{32} .

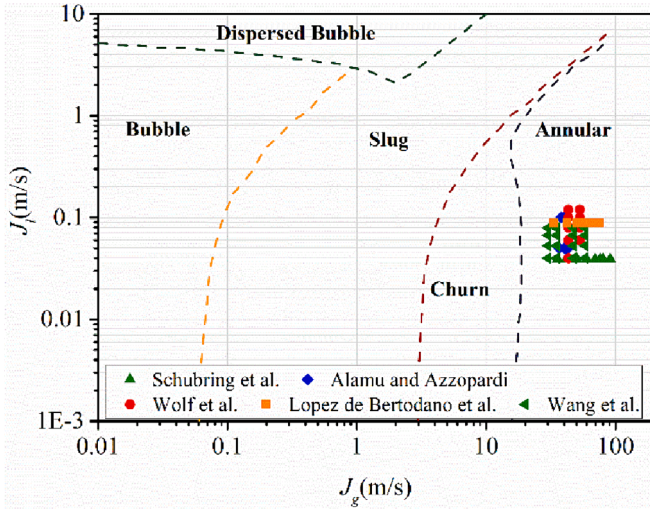


Fig. 8. Data points marker on a multiphase flow regime diagram (Barnea, 1987).

improve clarity, both the experimental and predicted liquid film thicknesses were magnified 10 times. The comparison results in Fig. 11 demonstrate the accuracy of the TLFM.

Fig. 12 depicts the evolution of the annular flow entering the 90° bend. The 90° bend is the first half of a U-bend; therefore, the evolution of the annular flow entering the 90° bend should be similar to that of the annular flow entering the first half of the U-bend discussed in Section 4.1. However, the curvature radius of the 90° bend is much longer than that of the U-bend. That is, the influence of the flow field boundary change caused by the bend is insignificant, greatly inhibiting the rotation of the gas phase entering the 90° bend. The liquid phase process in the liquid film flowing from the outside to the inside and accumulating inside is also inhibited. As shown in Fig. 12, the increase in the liquid film thickness on the inside of the bend is not obvious. The diameter of the 90° bend is significantly less than that of the U-bend in Section 4.1. In the small-scale flow field, the effect of surface tension is more obvious, which can better maintain the annular flow shape entering the bend from the vertical straight section. As shown in Fig. 12, the evolution process of the annular flow entering the 90° bend is similar to that shown in Fig. 10. A droplet void is found at the region with an azimuth angle ϕ of approximately 30°. However, the change in the annular flow field is weaker than that in the U-bend based on the two above-mentioned reasons. The effect of gravity is not noticeable in this case, as well. The green dotted lines in Fig. 11 represent the calculated liquid film thickness without considering gravity.

4.3. Annular flow in a T-junction

Conte and Azzopardi (2003) experimentally studied the division of the annular flow at a horizontal T-junction. To do this, they used conductivity techniques to measure the film thickness on the circumference within the T-junction. Both the main and side branches of the T-junction had a 0.127 m diameter. The T-junction was placed horizontally. The main and side branches were on the same horizontal plane. Table 1 presents the experimental parameters. The TLFM is used herein to analyze the complex variation of the annular flow in the T-junction. The structured grids are used to mesh the computational domain, which is divided into injection and development regions, as shown in Fig. 4(d). The grid independence is verified by setting the liquid film thickness as the target variable, in the same way as discussed in Section 3.2.

4.3.1. Validation of the liquid film thickness

The liquid film thicknesses on the three cross-sections taken from the inlet, outlet, and branch of the T-junction are compared with the predicted results. Fig. 13 shows a comparison between the predicted and measured liquid film thicknesses, where the MAPE of the predictions is 35.99 %. Both the experimental and predicted liquid film thicknesses are magnified five times for clarity. The polar angles $\phi = 0^\circ$ and 180° denote the top and bottom of the T-junction, respectively. On the inlet and outlet cross-sections, the predicted liquid film thicknesses agree with the

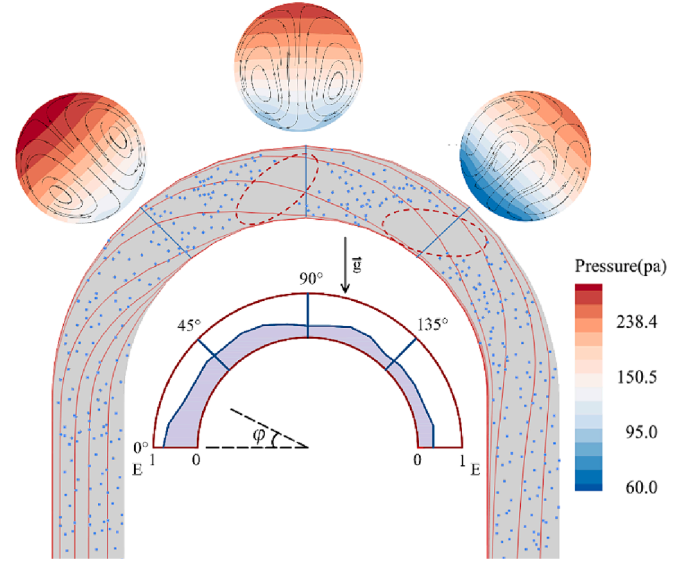


Fig. 10. Annular flow evolution in the U-bend.

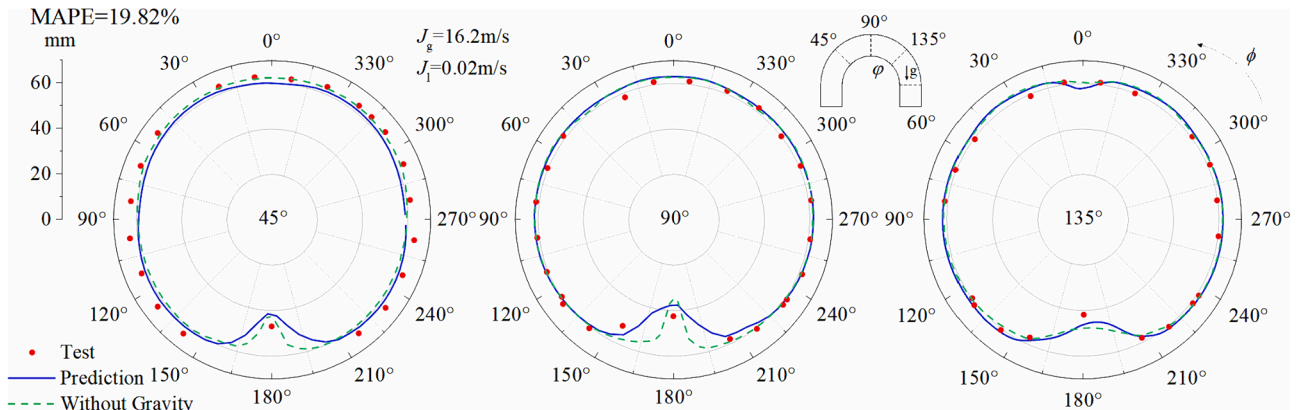


Fig. 9. Comparison between the predicted and measured liquid film thicknesses in a U-bend.

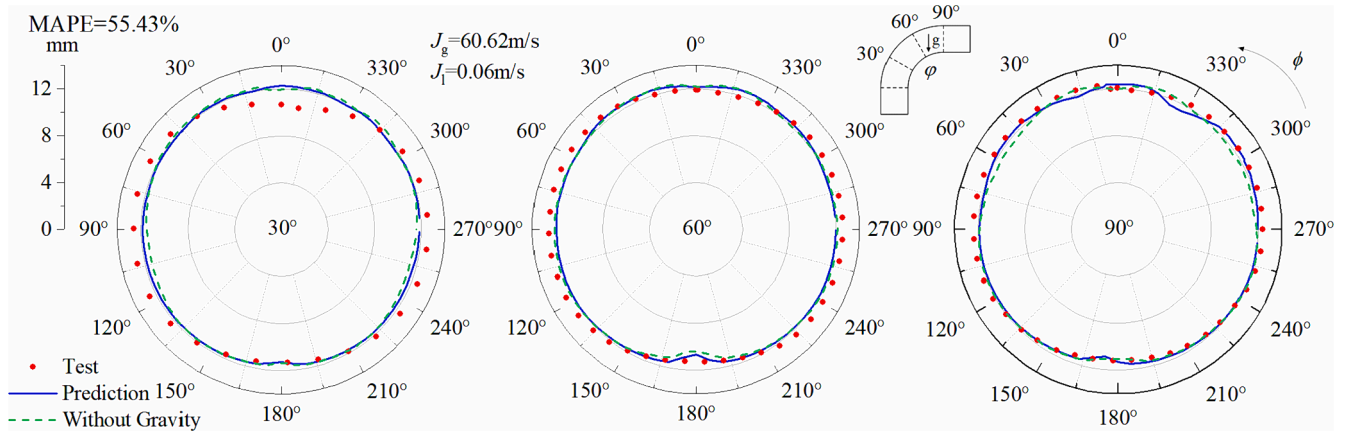


Fig. 11. Comparison of the predicted and measured liquid film thicknesses in a 90° bend.

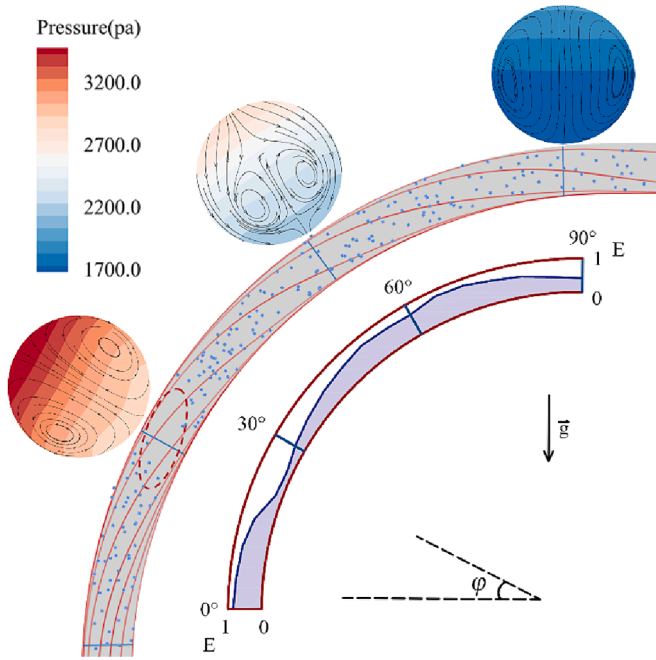


Fig. 12. Evolution of the annular flow in the 90° bend.

measured results at most locations, except for the bottom of the sections. Both the predicted and measured results show that the liquid film abruptly thickens at the bottom of the sections. However, the predicted thickest liquid film is thinner than that in the experimental data. On the cross-section in the side branch, the experimental and predicted results illustrate the same distribution of the liquid film thickness. That is, the liquid phase is almost evenly distributed on the pipe circumference. The liquid film thickness slightly increases near the polar angles of 120° and 240°. Similarly, the predicted thickest liquid film is thinner than the measured results.

The reason discussed in Section 3.3 can explain the deviation between the measured and predicted results. Droplets are deposited on the liquid film at the bottom of the pipe by gravity, causing the gas phase to be mixed into the liquid film. The TLFM assumes that the liquid film consists only of the liquid phase; hence, the predicted liquid film is thinner than the experimental result where the deposition process violently occurs.

4.3.2. Flow field evolution in the T-junction

Fig. 14 shows the evolution of the annular flow entering the T-junction. Cross-sections A, B and C are taken from the inlet, side branch and outlet of the T-junction, respectively. The black arrow and red solid lines along the T-junction represent the streamlines of the gas phase on the symmetry plane of the gas core and the liquid phase in the liquid film, respectively. The blue scatter points are the droplets near the corresponding cross-sections.

Accordingly, we set up three observation regions on the inlet, outlet, and side sections of the T-junction to record the flow rates of the droplets and liquid film. Calculating the droplet flow rate requires searching all

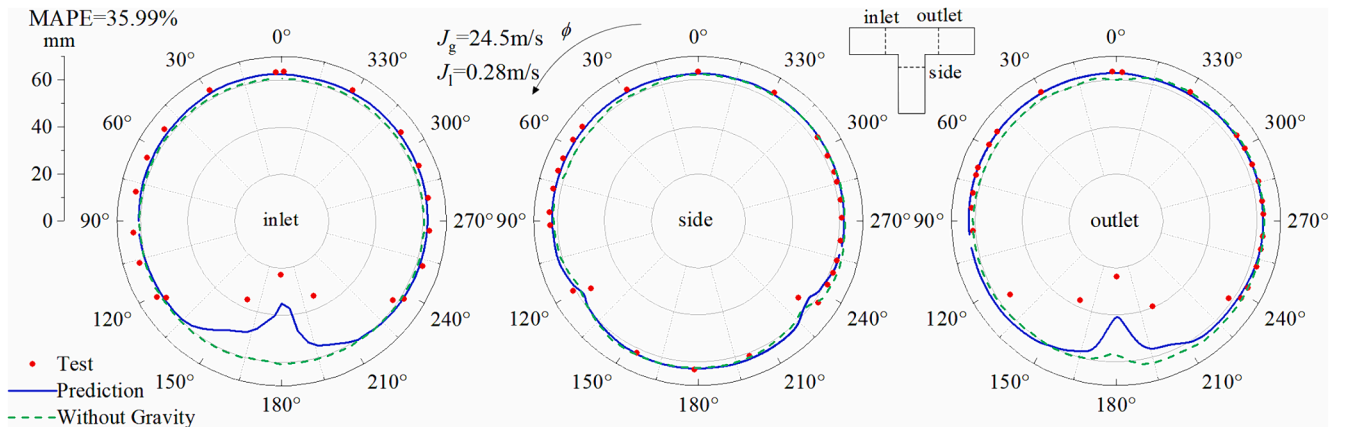


Fig. 13. Comparison between the predicted and measured liquid film thicknesses in a T-junction.

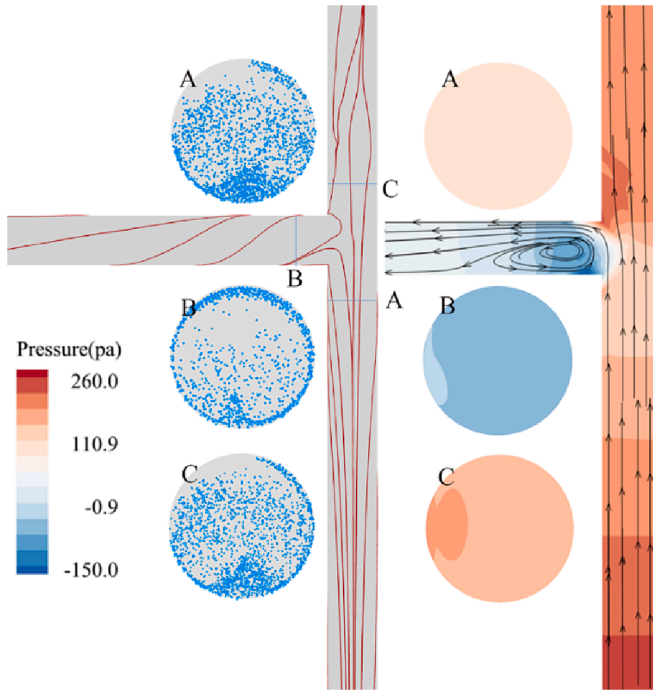


Fig. 14. Evolution of the annular flow in the T-junction.

droplets in the observation region. As shown in Fig. 2, the observation region contained N cell layers, and the length of each cell layer is Δl . Let us now suppose that n_d parcels are found in the j th cell layer, and the axial velocity of parcel i is v_{dij}^z . The droplet flow rate in the j th cell layer is obtained as follows:

$$w_{dj} = \sum_{i=1}^{n_d} \frac{v_{dij}^z M_{dij}}{\Delta l} = \sum_c \frac{v_{dij}^z M_{dij}}{\Delta l} \quad (25)$$

The average droplet flow rate in the observation region is calculated as

$$w_d = \frac{1}{N} \sum_z \left(\sum_c \frac{v_{dij}^z M_{dij}}{\Delta l} \right) = \sum_z \sum_c \frac{v_{dij}^z M_{dij}}{L_{data}} \quad (26)$$

The upstream of the T-junction is a horizontal straight pipeline, wherein the liquid phase accumulated at the pipe bottom. The green dotted lines in Fig. 13 represent the calculated film thickness without considering gravity. The liquid film is almost evenly distributed on the pipe circumference. Different from the annular flow discussed in Sections 4.1 and 4.2, gravity plays a key role in forming the annular flow in the horizontal pipeline. As shown in cross-section A, droplets gather toward the pipe bottom due to gravity. Some of the droplets are deposited in the liquid film at the bottom of the pipe, causing the local liquid film to thicken. The liquid phase in the liquid film also flows to the pipe bottom, which increases the liquid film thickness. Gravity shapes the annular flow through the two abovementioned actions, resulting in the thickest liquid film at the pipe bottom.

When the annular flow in the horizontal pipe passes through the T-junction, both the liquid film and the droplets are split through the main and side branches. Fig. 14 depicts the gas phase streamlines on the symmetrical surface of the T-junction. Most of the gas phase moved along the main branch. Only the gas phase close to the side arm enters the branch and forms a cyclone. Due to the gas phase influence, most of the droplets continue to move along the main branch, and a small part enter the side branch. In the present case, 4.539 % of the liquid phase enters the T-junction in droplet form. The proportions of the droplets in the main and side branches to the total liquid phase are 2.915 % and 0.0395 %, respectively. This implies that 1.585 % of the liquid phase

changes from droplets to the liquid film when flowing through the T-junction. In the main branch, the gas and liquid phases maintain the flow pattern of the upstream flow field, that is, the liquid phase accumulates at the pipe bottom. The droplet distribution on cross-section C are quite similar to that on cross-section A. In the side branch, the liquid film accumulating at the bottom is blown away by the cyclone. Also affected by the cyclone, the droplets near cross-section B gather around the pipe wall rather than towards the bottom. The liquid film is thickest near the polar angles of 120° and 240°. The liquid film proportions in the main and side branches to the total liquid phase are 86.082 % and 10.963 %, respectively. Overall, 88.997 % of the liquid phase continues to move along the main branch after passing through the T-junction.

5. Conclusions

The TLFM is applied to investigate the annular flow in the U-bend, 90° bend, and T-junction. Before its application, the TLFM is thoroughly verified by comparison with the experimental data from the tests performed in straight pipes. Several conclusions can be drawn from the study:

- (1) In the U-bend, the droplets affect the outside of the bend due to inertia and merge into the local liquid film. Upon entering the bend, the gas phase forms two symmetrical cyclones that drive the liquid film to flow from the outside to the inside of the bend. The liquid phase in the liquid film tends to accumulate on the inside of the bend due to gravity. The two symmetrical cyclones generated by the gas phase dominate the evolution of the annular flow entering the bend.
- (2) In the 90° bend, the mechanism of the annular flow evolution is similar to that in the U-bend. However, the bend curvature is much larger than that of the U-bend, resulting in a less drastic change in the annular flow morphology. The smaller pipe diameter of the 90° bend makes surface tension more effective in maintaining the annular flow shape formed in the straight pipe section.
- (3) In the T-junction, gravity plays a dominant role in the annular flow formation in the upstream pipeline. After entering the T-junction, most of the droplets and the liquid phase in the liquid film move along the main branch. The cyclone formed at the side branch blows away the liquid phase accumulating at the pipe bottom.

The current TLFM cannot account for the presence of gas mixed in the liquid film, leading to a certain degree of deviation between predicted and measured results. The TLFM needs to be modified to address this issue. Additionally, further research should aim to integrate the breakup and coalescence processes of droplets, as well as consider the heat transfer between the gas and liquid phases.

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CRediT authorship contribution statement

Ri Zhang: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Mengyan Ding:** Data curation, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Abdulkadir, M., Azzi, A., Zhao, D., Lowndes, I.S., Azzopardi, B.J., 2014. Liquid film thickness behaviour within a large diameter vertical 180° return bend. *Chem. Eng. Sci.* 107, 137–148.
- Adechy, D., Issa, R.I., 2004. Modelling of annular flow through pipes and T-junctions. *Comput. Fluids* 33 (2), 289–313.
- Alamu, M.B., Azzopardi, B.J., 2011. Simultaneous investigation of entrained liquid fraction, liquid film thickness and pressure drop in vertical annular flow. *J. Energy Resour. Technol.* 133 (2), 023103.
- Alipchenkov, V.M., Nigmatulin, R.I., Soloviev, S.L., Stonik, O.G., Zaichik, L.I., Zeigarnik, Y.A., 2004. A three-fluid model of two-phase dispersed-annular flow. *Int. J. Heat Mass Transfer* 47 (24), 5323–5338.
- Aliyu, A.M., Almagbrok, A.A., Baba, Y.D., Lao, L., Yeung, H., Kim, K.C., 2017. Upward gas-liquid two-phase flow after a U-bend in a large-diameter serpentine pipe. *Int. J. Heat Mass Transfer* 108, 784–800.
- Ambrosini, W., Andreussi, P., Azzopardi, B.J., 1991. A physical based correlation for drop size in annular flow. *Int. J. Multiphase Flow* 17, 497–507.
- Ansys, 2018. *Fluent. Theory Guide*, ANSYS Inc, Canonsburg, PA.
- Barnea, D., 1987. A unified model for predicting flow-pattern transition for the whole range of pipe inclinations. *Int. J. Multiphase Flow* 13 (1), 1–12.
- Cioncolini, A., Thome, J.R., 2012. Entrained liquid fraction prediction in adiabatic and evaporating annular two-phase flow. *Nucl. Eng. Des.* 243, 200–213.
- Conte, G., Azzopardi, B.J., 2003. Film thickness variation about a T-junction. *Int. J. Multiphase Flow* 29, 305–328.
- Fore, L.B., Beus, S.G., Bauer, R.C., 2000. Interfacial friction in gas-liquid annular flow: analogies to full and transition roughness. *Int. J. Multiphase Flow* 26, 1755–1769.
- Fukano, T., Furukawa, T., 1998. Prediction of the effects of liquid viscosity on interfacial shear stress and frictional pressure drop in vertical upward gas-liquid annular flow. *Int. J. Multiphase Flow* 24, 587–603.
- Ghosh, S., Das, G., Das, P.K., 2011. Simulation of core annular in return bends – A comprehensive CFD study. *Chem. Eng. Res. Des.* 89 (11), 2244–2253.
- Gosman, A.D., Ioannides, E., 1983. Aspects of computer simulation of liquid-fueled combustors. *J. Energy* 7 (6), 482–490.
- Govier, G.W., Aziz, K., 1972. *The Flow of Complex Mixtures in Pipes*. Van Nostrand Reinhold, New York.
- Guan, X., Zhao, Y., Wang, J., Zhang, D., 2015. Numerical analysis of quasi-steady flow characteristics in large diameter pipes with low liquid loading under high pressure. *J. Nat. Gas Sci. Eng.* 26, 907–920.
- Han, H.W., Gabriel, K., 2007. A numerical study of entrainment mechanism in axisymmetric annular gas-liquid flow. *J. Fluids Eng.* 129 (3), 293–301.
- Henstock, W.H., Hanratty, T.J., 1976. The interfacial drag and the height of the wall layer in annular flows. *AIChE J.* 22 (6), 990–1000.
- Hewitt, G.F., Govan, A.H., 1990. Phenomenological modelling of non-equilibrium flows with phase change. *Int. J. Heat Mass Transfer* 33 (2), 229–242.
- Hewitt, G.F., Hall-Taylor, N.S., 1970. *Annular Two-phase Flow*. Pergamon Press, Oxford.
- Ishii, M., Mishima, K., 1984. Two-fluid model and hydrodynamics constitutive relations. *Nucl. Eng. Des.* 82, 107–126.
- Jiang, W., Wu, J., Du, S., Liu, C., Liu, Y., 2019. CFD simulation on oil-water annular flow through CAF generator device. *Energy Sources, Part a: Recovery. Util. Environ. Eff.* 42 (15), 1861–1873.
- Katakka, I., Ishii, M., Mishima, K., 1983. Generation and size distribution of droplet in annular two-phase flow. *J. Fluids Eng.* 105 (2), 230–238.
- Kataoka, I., Ishii, M., Nakayama, A., 2000. Entrainment and deposition rates of droplets in annular two-phase flow. *Int. J. Heat Mass Transfer* 43, 1573–1589.
- Kiran, R., Ahmed, R., Salehi, S., 2020. Experiments and CFD modelling for two phase flow in a vertical annulus. *Chem. Eng. Res. Des.* 153, 201–211.
- Kishore, B.N., Jayanti, S., 2004. A multidimensional model for annular gas-liquid flow. *Chem. Eng. Sci.* 59, 3577–3589.
- Li, H.P., Anglart, H., 2016. Prediction of dryout and post-dryout heat transfer using a two-phase CFD model. *Int. J. Heat Mass Transfer* 99, 839–850.
- Li, H.P., Anglart, H., 2019. Dryout prediction with CFD model of annular two-phase flow. *Nucl. Eng. Des.* 349, 20–26.
- Li, H., Li, J., Zhou, Y., Liu, B., Sun, B., Yang, D., Fan, X., 2017. Phase split characteristics of slug and annular flow in a dividing micro-T-junction. *Exp. Therm. Fluid Sci.* 80, 244–258.
- Ligrani, P.M., Moffat, R.J., 1986. Structure of transitionally rough and fully rough turbulent boundary layers. *J. Fluid Mech.* 162, 69–98.
- Liu, Y., Cui, J., Li, W., 2011. A two-phase, two-component model for vertical upward gas-liquid annular flow. *Int. J. Heat Fluid Flow* 32, 796–804.
- Liu, H., Liu, W., Yan, P., Chen, D., Dong, K., Qin, J., Liu, H., 2022. The role mechanism of vapor-liquid behavior on boiling crisis triggering. *Int. J. Heat Mass Transfer* 196, 123248.
- Liu, H., Dong, K., Yan, P., Zhao, J., Hibiki, T., Qin, J., Liu, H., Chen, D., 2024. Experimental and theoretical study on liquid-vapor behaviour characteristics near CHF. *Int. J. Therm. Sci.* 198, 108879.
- Lopez de Bertodano, M.A., Jan, C.S., Beus, S.G., 1997. Annular flow entrainment rate experiment in a small vertical pipe. *Nucl. Eng. Des.* 178 (1), 61–70.
- Maddock, C., Lacey, P.M.C., Patrick, M.A., 1974. The structure of two-phase flow in a curved pipe. *Symposium on Multiphase Flow Systems*, Institution of Chemical Engineers Symposium, Glasgow, UK, Paper, p. J2.
- Mayer, E., 1961. Theory of liquid atomization in high velocity gas streams. *ARS J.* 31, 467–473.
- Ribeiro, A.M., Bott, T.R., Jepson, D.M., 2001. The influence of a bend on drop sizes in horizontal annular two-phase flow. *Int. J. Multiphase Flow* 27 (4), 721–728.
- Savant, P., Ishii, M., Mori, M., 2008. Droplet entrainment correlation in vertical upward co-current annular two-phase flow. *Nucl. Eng. Des.* 238 (6), 1342–1352.
- Schadel, S.A., Leman, G.W., Binder, J.L., Hanratty, Y.J., 1990. Rates of atomization and deposition in vertical annular flow. *Int. J. Multiphase Flow* 16 (3), 363–374.
- Schubring, D., Ashwood, A.C., Shedd, T.A., Hurlburt, E.T., 2010a. Planar laser-induced fluorescence (PLIF) measurements of liquid film thickness in annular flow. Part I: Methods and data. *Int. J. Multiphase Flow* 36 (10), 815–824.
- Schubring, D., Shedd, T.A., Hurlburt, E.T., 2010b. Planar laser-induced fluorescence (PLIF) measurements of liquid film thickness in annular flow. Part II: Analysis and comparison to models. *Int. J. Multiphase Flow* 36 (10), 825–834.
- Stanton, D.W., Rutland, C.J., 1998. Multi-dimensional modeling of thin liquid films and spray-wall interactions resulting from impinging sprays. *Int. J. Heat Mass Transfer* 41 (20), 3037–3054.
- Tatterson, D.F., Dallman, J.C., Hanratty, T.J., 1977. Drop sizes in annular gas-liquid flows. *AIChE J.* 23, 68–76.
- Wallis, G.B., 1969. *One-dimensional Two-phase Flow*. McGraw-Hill, New York.
- Wang, Z., Liu, H., Zhang, Z., Sun, B., Zhang, J., Lou, W., 2020. Research on the effects of liquid viscosity on droplet size in vertical gas-liquid annular flows. *Chem. Eng. Sci.* 220, 115621.
- Whalley, P.B., 1987. *Boiling, condensation, and gas-liquid flow*. Oxford University Press, New York.
- Wolf, A., Jayanti, S., Hewitt, G.F., 2001. Flow development in vertical annular flow. *Chem. Eng. Sci.* 56 (10), 3221–3235.
- Wu, J., Jiang, W., Liu, Y., He, Y., Chen, J., Qiao, L., Wang, T., 2020. Study on hydrodynamic characteristics of oil-water annular flow in 90° elbow. *Chem. Eng. Res. Des.* 153, 443–451.
- Zahedi, R., Rad, A.B., 2022. Numerical and experimental simulation of gas-liquid two-phase flow in 90-degree elbow. *Alex. Eng. J.* 61, 2536–2550.
- Zhang, R., Zhang, S.S., Ding, M.Y., 2024. Modelling of annular flow and sand erosion in bends using a thin liquid film method. *Powder Technol.* 432, 119111.